

Questioning the consensus and paths forward

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Current mainstream Business cycle analysis

background

- ▶ Much of the mainstream business cycle analysis, such as done in Central banks, works off the RBC model augmented with sticky prices: ie, the New Keynesien model.
- ▶ Some take the model seriously quantitatively, other use it more for qualitative analysis (often 3 equation version without capital)
- ▶ Much of the current work in the area involves extending the model richness on the financial side, and addressing issue of zero lower bound.
- ▶ This is active area of research.
- ▶ Here we want to question this research agenda: claim: may be desirable to rebuild the skeleton
- ▶ Some background first.

The NK consensus

background

- ▶ Keynesian macro was not primarily set up as a theory of cycles. Keynes was primarily motivated by explaining what to do when in a recession/depression, as opposed to explaining why they happen. Idea: Need to stimulate demand.
- ▶ Introduced the notion of deficient demand. However, definition not perfectly clear.(A tentative definition related to fiscal policy?)
- ▶ Source of deficient demand controversial from beginning: absence of price adjustment versus coordination problem.
- ▶ The NK put the model on optimization basis, followed Hicks by emphasizing sticky nominal price as why demand could be deficient. This places monetary policy at center.

Questioning the consensus

- ▶ In some respects, the NK model can be seen as very close to monetarism since, if monetary policy can be set appropriately, model suggests that the economy should function rather well or at least not exhibit deficient demand. Reason: sticky prices is the central market failure causing deficient demand. (without sticky nominal prices, outcomes may not be optimal, but the essence of deficient demand in the model often disappears)
- ▶ An attractive feature of the NK model is that it is a framework which allows us to interpret or decompose the cause of cycles.

Questioning the consensus

- ▶ Viewed as a decomposition method, part of its validity of the NK approach comes from the believability of what it delivers
- ▶ Strengths: (1) can reproduce the effects of monetary shock (main test) , (2) fits the data well
- ▶ potential weaknesses
- ▶ Findings:
 1. Generally explains recessions as resulting from the co-incidence of many negative shocks
 2. Relies on many hard to interpret shocks (ex: markup shocks)
 3. Suggests business cycle properties should change significantly as we change monetary policy
 4. Difficulty explaining situations like Japan
 5. Unclear if capable of explaining changing recent behavior of macro aggregates (change in frequency as opposed to great moderation)

Questioning the consensus

Way forward

- ▶ It is often claimed that the NK model offers a demand driven theory of the BC. Does it? . Not clear. Demand shocks often found to have a somewhat secondary role, especially for labor. Mark-up shocks often most important for labor.
- ▶ Gives rise to question: what is meant by demand shocks? Common answer: monetary shocks and government spending. But not very compelling, both for conceptual reasons and for quantitative reasons. Other possible candidates: shocks that act to change agent's incentive to go out and purchase good, related to notion of optimism and pessimism (rational or irrational).

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Way forward

- ▶ Two reasons why demand shocks often play relatively modest role in standard NK. Minor reason. Standard model do not always incorporate many of such shocks. More important. given the framework, such shocks are – for structural reasons– unlikely to play big role and they would either need to produce negative co-movements, or create pro-cyclical inflation; neither of which is much present in the data.
- ▶ Implication: either (1) model is right, and demand plays much less of a role than often thought or (2) core of model is flawed and this is biasing inference on the role of demand in the BC

Questioning the consensus

- ▶ Way forward: Examine what is limiting the standard business cycle model from generating demand driven business cycles in the absence of sticky prices. Then explore extensions/modification and evaluate their relevance.
- ▶ Ultimate objective: explore alternative macro models where demand can have more effect, and see whether this gives a better narrative of the (boom-bust) cycle.
- ▶ Three narratives we find compelling are compelling. Optimism and pessimism driven cycles (news, noise, risk, ambiguity, sentiment), shocks to the financial market drives the business cycle, and the economy may be inherently unstable and endogenously may favor cycles. . Here focus on first and third, while aware that second may also be relevant.

Questioning the consensus

Way forward

- ▶ Closely related issue: **liquidations**. When booms are driven by demand, this means they finish with 'high' capital stocks. In which case, recession may reflect the liquidation of such stocks. Question: Is this an important component in understanding cycles?

Questioning the consensus

Way forward

► Steps

1. Briefly discuss candidate demand shocks.
2. Give simple example of news-noise boom bust cycles, with liquidation playing a role
3. Look at Var evidence in support of such shocks and mechanisms
4. Overview different setups where demand shock can produce BC phenomena within a Walrasian framework
5. . Consider more substantial departures from Walrasian framework, emphasizing coordination failures, which may better capture notion of deficient demand.
 - 5.1 Hopefully be able to use framework for understanding issues such as secular deficient demand like in Japan, differences in cycles between emerging and developed economies, international business cycles.

Next steps

- ▶ Our general long term research agenda: Explore different mechanisms which may explain how demand shocks can cause business type phenomena in absence of sticky prices, embed in full model with sticky prices (since need to be consistent with evidence on monetary shocks), evaluate general performance as compared with NK model.
- ▶ Challenge: great reluctance – for good reasons–of changing (beside tweaks) the current, internally consistent, tractable, NG framework on which both the RBC and NK model build. Hard to beat a model which has been extended to fit the data.
- ▶ Potentially important element: thinking of capital in terms of household capital versus firm capital. Different implications for the marginal product of labor in the market.

Potential Demand shocks

- ▶ Searching for why agents repeatedly go through periods where they are more ready and willing to take risks and invest (either in durable goods , houses and iphysical investment), and subsequently periods where they do appear unwilling to do so, and this behavior leads to fluctuations.
- ▶ A set of such shocks suggested the the literature
 1. News and Noise
 2. Uncertainty
 3. Ambiguity
 4. sentiment
- ▶ Begin by focusing on the first, as we know it best.